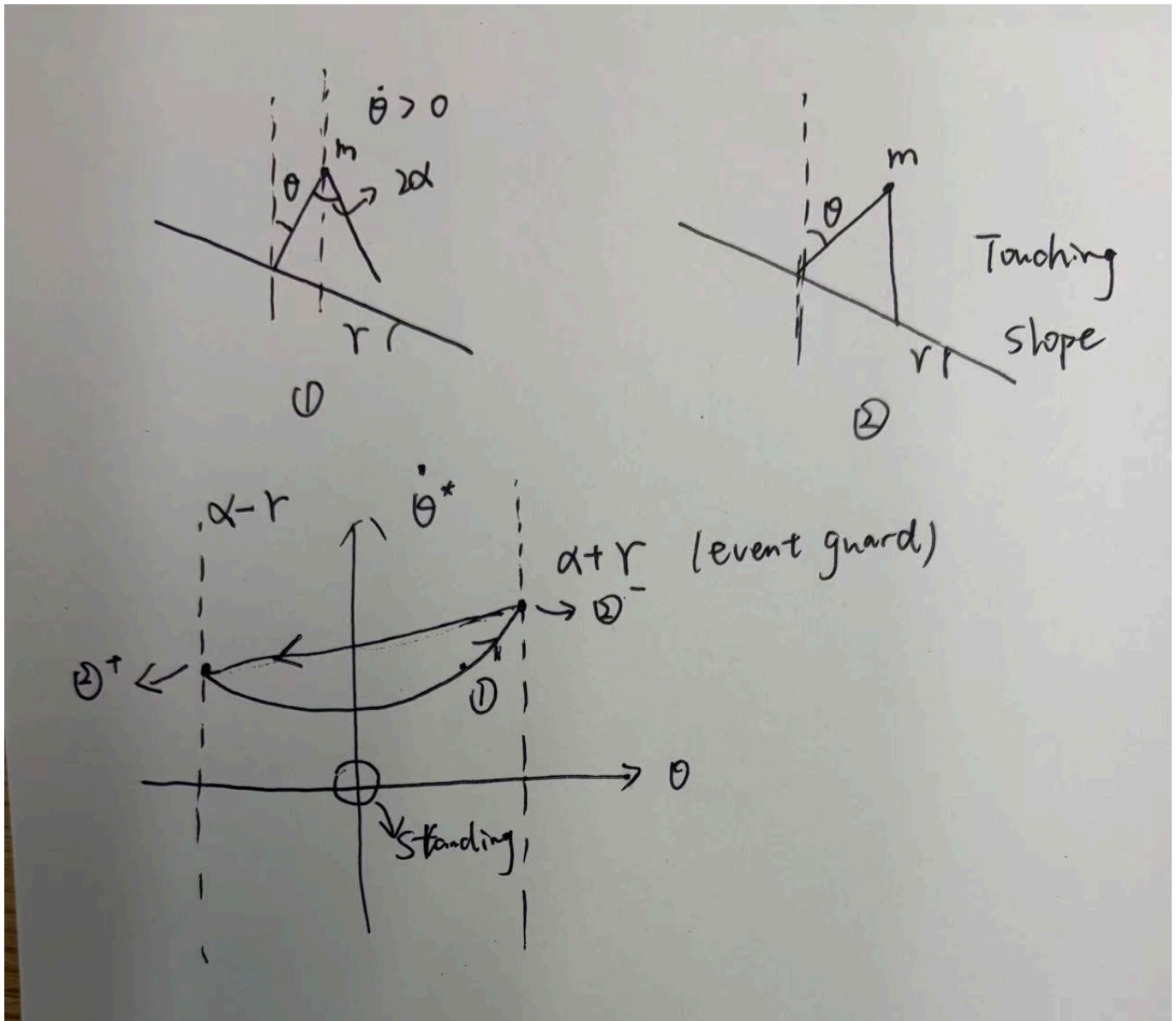


Assignment 2 Report

To run the codes for assignment 2, please use:

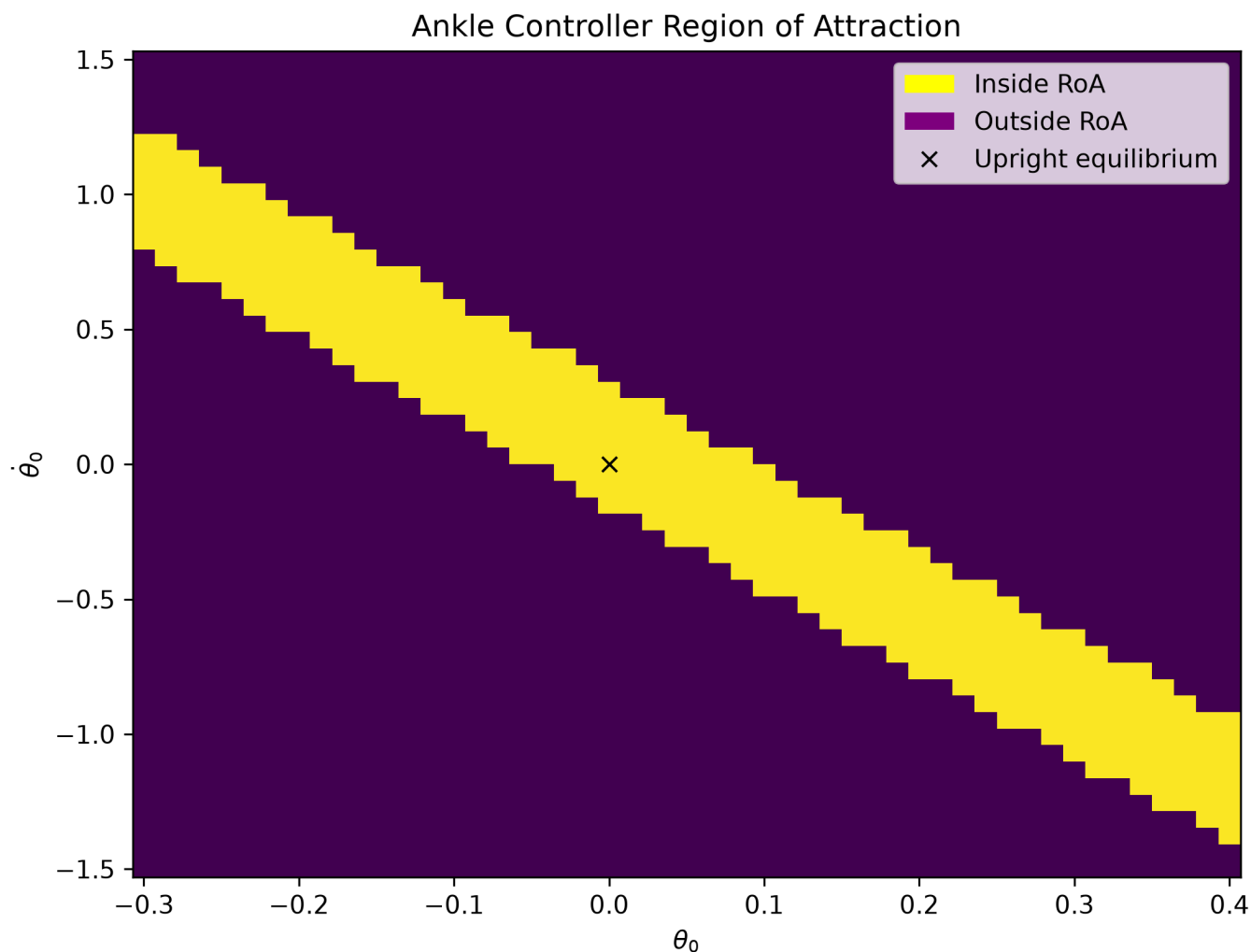
```
uv run python assignment_2.py
```

Sketch



Region of attraction

The saturated feedback-linearizing ankle controller stabilizes the upright state inside the sampled region below.



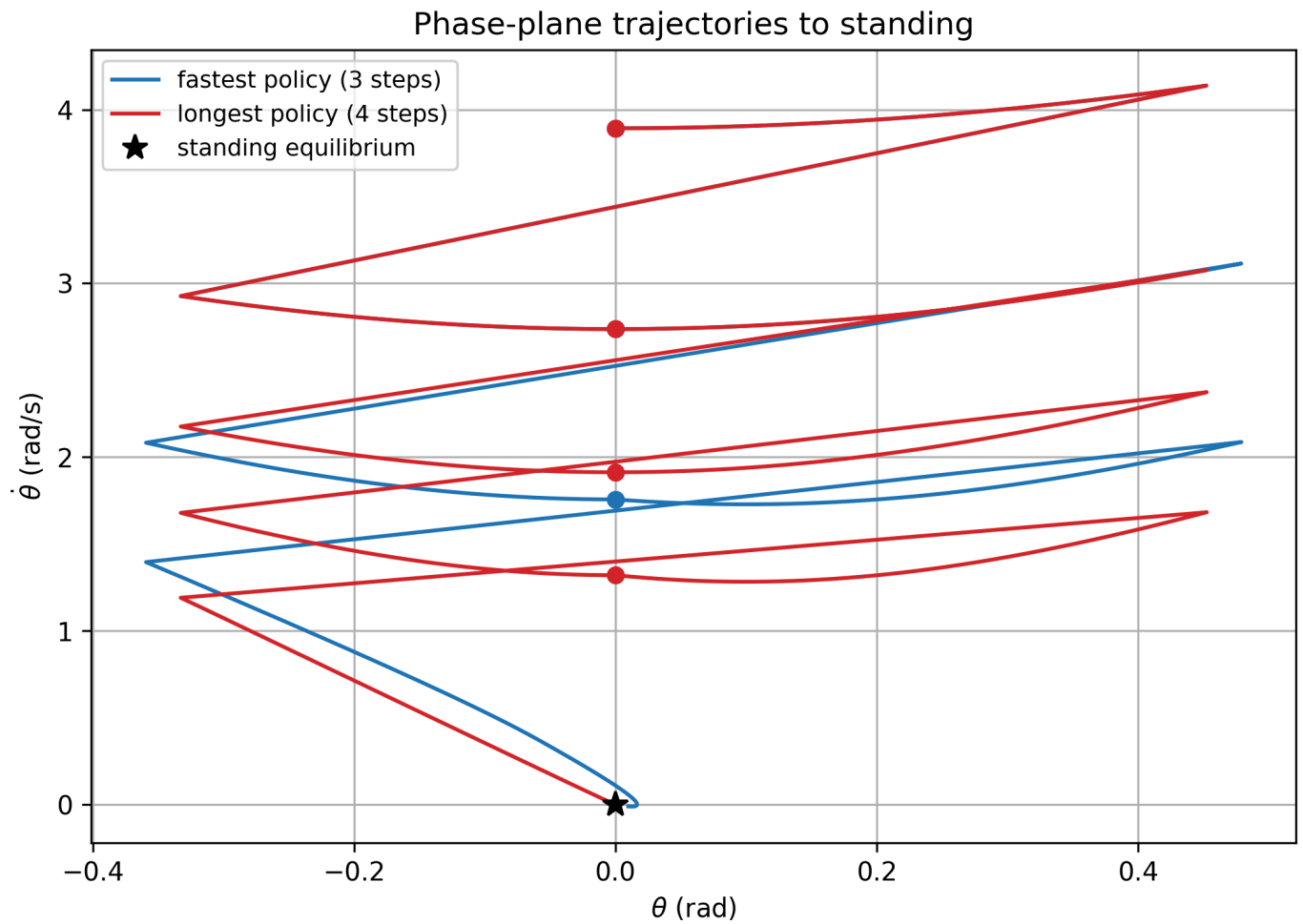
Poincare section and grid

I use the upward crossing of $\theta = 0$ with $\dot{\theta} > 0$. It is transverse to the walking orbit and leaves $\dot{\theta}$ as the scalar return-map state. The final lookup table uses 100 velocity points and 20 landing-angle points.

I choose 100 for θ_0 , because it gives an relative error rate of less than 0.5%, while other smaller desolutions don't. And for the choice of α , I just simply set it to a constant, for it does not influence the dynamics except for the time when touching the slope.

Three-step trajectory and longest walk

The selected initial condition is $\theta_0 = 0$ and $\dot{\theta}_0 = 3.893$ rad/s. The lookup policy reaches the standing RoA in three steps. Enumerating every valid action from this initial grid cell gives a maximum finite duration of four steps before entering the RoA; the figure displays the minimum-step (blue) and maximum-duration (red) policies.



Steps to standstill

The plot below gives the minimum number of footsteps to enter the standing RoA for each initial velocity on the Poincare section. The red star marks the three-step initial condition above.

